

Verification of Theorem 1

Summary of `verify_theorem1.py`

(a) Theory

The bivariate generating function

The paper introduces the bivariate generating function

$$H(y, q) = \sum_{r \geq 0} h_r(q) y^r,$$

where $h_r(q)$ is the generating function for teeth of base length r .

Theorem 1 gives an explicit closed-form expression for $H(y, q)$, namely

$$H(y, q) = \frac{1}{(yq; q)_\infty} - \sum_{m \geq 1} \frac{yq^m}{(yq; q)_m^2}.$$

This representation replaces the recursive definition of the $h_r(q)$ with a single analytic expression involving q-Pochhammer symbols.

Functional equation

The proof of Theorem 1 is based on the functional equation

$$H(y, q)(1 - qy)^2 = H(qy, q) - qy.$$

This equation encapsulates the recurrence structure of the generating functions and uniquely characterises $H(y, q)$.

Recovery of the individual generating functions

Since

$$H(y, q) = \sum_{r \geq 0} h_r(q) y^r,$$

the coefficient of y^r in the closed-form expression must reproduce $h_r(q)$.

In particular,

$$[y]H(y, q) = h_1(q), \quad [y^2]H(y, q) = h_2(q).$$

These provide immediate consistency checks on the theorem.

(b) What the script does

The script performs three independent verifications of Theorem 1.

Check A: Functional equation

It constructs a truncated series representation of $H(y, q)$ from the recurrence-generated functions $h_r(q)$ and verifies that

$$H(y, q)(1 - qy)^2 = H(qy, q) - qy$$

holds symbolically through order y^6 . This check tests the recurrence-generated series directly; the closed-form formula of Theorem 1 does not appear in Check A. The verification establishes that the truncated recurrence series satisfies the functional equation that characterises $H(y, q)$, which is a necessary condition for the theorem.

Check B: Coefficient extraction

It extracts the coefficients

$$[y]H(y, q) \quad \text{and} \quad [y^2]H(y, q)$$

directly from the closed-form formula and verifies that they reproduce the known generating functions $h_1(q)$ and $h_2(q)$.

Check C: Numerical evaluation

It evaluates $H(y, q)$ at

$$(y, q) = (0.3, 0.5)$$

using two completely different methods:

1. Summation of the recurrence-generated coefficients $h_r(q)$;
2. Direct evaluation of the closed-form expression from Theorem 1.

The resulting numerical values are compared, with agreement declared if the absolute difference is less than 10^{-9} .

(c) How the script does it

Construction of $h_r(q)$

`compute_h(max_r)` generates the functions recursively using

$$(1 - q^r)h_r(q) = 2q h_{r-1}(q) - q^2 h_{r-2}(q).$$

The resulting rational functions are simplified symbolically using SymPy.

Check A: Functional equation

The script forms the truncated generating function

$$H(y, q) = \sum_{r=0}^6 h_r(q) y^r,$$

which includes the y^0 term $h_0 = 1$ — the boundary condition $H(0, q) = 1$ used in the proof of Theorem 1. It then computes

$$H(y, q)(1 - qy)^2$$

and

$$H(qy, q) - qy,$$

expands both as polynomials in y , and subtracts them.

The theorem is verified if the resulting polynomial is identically zero.

Check B: Symbolic coefficient extraction

Rather than performing general automated coefficient extraction, the script encodes the specific logarithmic differentiation steps as hardcoded symbolic expressions. This is a symbolic confirmation of the manual calculations presented in Remark 2 of the paper (the consistency check for $r = 1$ and $r = 2$).

For the first term of Theorem 1,

$$\frac{1}{(yq; q)_{\infty}^2},$$

the script uses logarithmic differentiation and known q -series identities to obtain the coefficients of y and y^2 .

The corresponding coefficients of the subtraction term

$$\sum_{m \geq 1} \frac{yq^m}{(yq; q)_m^2}$$

are then evaluated separately.

Combining the two contributions yields symbolic expressions for

$$[y]H(y, q)$$

and

$$[y^2]H(y, q),$$

which are compared against the recurrence-generated values of $h_1(q)$ and $h_2(q)$.

Check C: Numerical evaluation

The script evaluates

$$H(0.3, 0.5)$$

in two independent ways.

Recurrence evaluation

The generating functions $h_r(0.5)$ are generated recursively up to $r = 39$, and

$$H(0.3, 0.5) = \sum_{r=0}^{39} h_r(0.5)(0.3)^r$$

is computed.

Closed-form evaluation

The product

$$\frac{1}{(yq; q)_{\infty}^2}$$

and the correction sum

$$\sum_{m \geq 1} \frac{yq^m}{(yq; q)_m^2}$$

are both truncated after 40 terms and evaluated numerically.

The two resulting approximations are then compared. At $(y, q) = (0.3, 0.5)$ the r -th term of the recurrence sum is of order $(0.3)^r$, which decays rapidly, so 40 terms is more than sufficient for the required precision. Agreement is declared if the absolute difference is less than 10^{-9} .

(d) Output produced

Functional equation verification

The script prints

$$H(y, q)(1 - qy)^2 - (H(qy, q) - qy),$$

expanded through order y^6 .

A successful verification gives

$$\text{LHS} - \text{RHS} = 0$$

confirming agreement term-by-term.

Coefficient checks

The script prints the extracted values of

$$[y]H(y, q)$$

and

$$[y^2]H(y, q),$$

alongside the independently generated functions $h_1(q)$ and $h_2(q)$.

For each coefficient it reports whether the symbolic expressions are identical.

Numerical comparison

The values

$$H(0.3, 0.5)$$

obtained from the recurrence and from Theorem 1 are displayed.

The script reports:

- the two numerical values;
- their absolute difference;
- whether their absolute difference is less than 10^{-9} .

(e) What we learn

Theorem 1 satisfies the defining functional equation. The symbolic verification confirms that the proposed closed-form expression obeys the same functional equation that characterises the generating function $H(y, q)$.

The formula reproduces the known low-order generating functions. The coefficients $[y]H$ and $[y^2]H$ recover $h_1(q)$ and $h_2(q)$ exactly, demonstrating consistency with the recurrence theory.

The analytic and recursive constructions agree numerically. The evaluation at $(y, q) = (0.3, 0.5)$ provides an independent numerical confirmation that the closed-form expression and the recurrence-generated series describe the same function.

The verification is independent in three different ways. Functional-equation testing, coefficient extraction, and numerical evaluation each probe different aspects of the theorem, making simultaneous agreement highly persuasive.

The script acts as a validation of Theorem 1 rather than a derivation. Its purpose is to confirm that the explicit q-series formula is consistent with the recurrence structure and the known generating functions, thereby providing computational support for the theorem presented in the paper.